# STM Exercise 5

Phase Diagrams

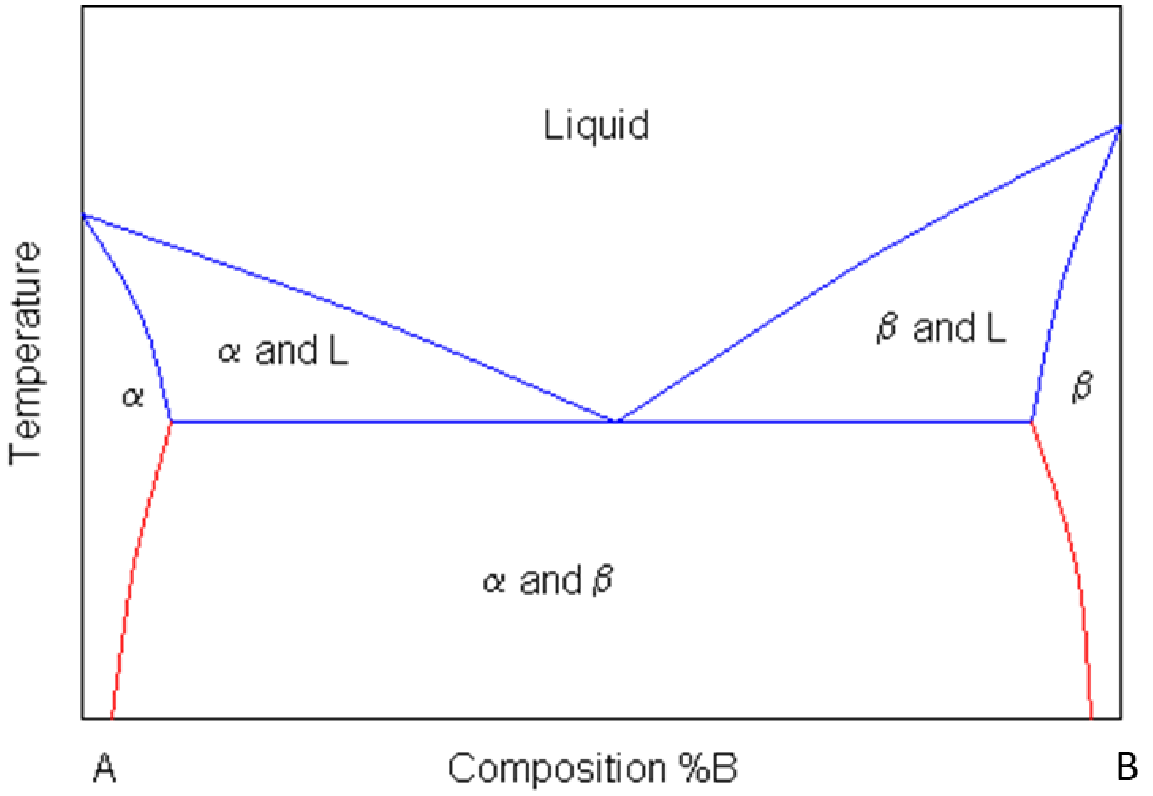
(full solubility in the solid state)

(no solubility in the solid state)

# Phase Diagram

it shows the equilibrium states of a mixture, so that given a temperature and composition, it is possible to calculate which phases will be formed, and in what quantities.

Where **L** stands for liquid, and **A** and **B** are the two components and **α and β** are two solid phases rich in A and B respectively.

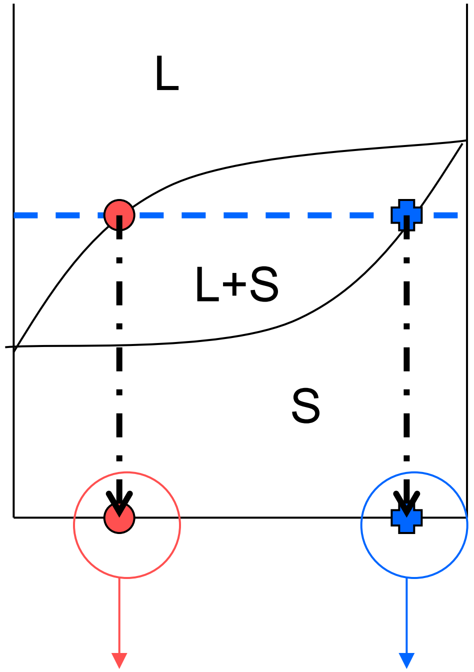



The blue lines represent the liquidus and solidus lines

The red lines involve a solid-to-solid transition (solvus lines)

## Determination of the compositions of phases in equilibrium in a biphasic zone:

## HORIZONTAL RULE




**Liquid composition**

**Solid composition**

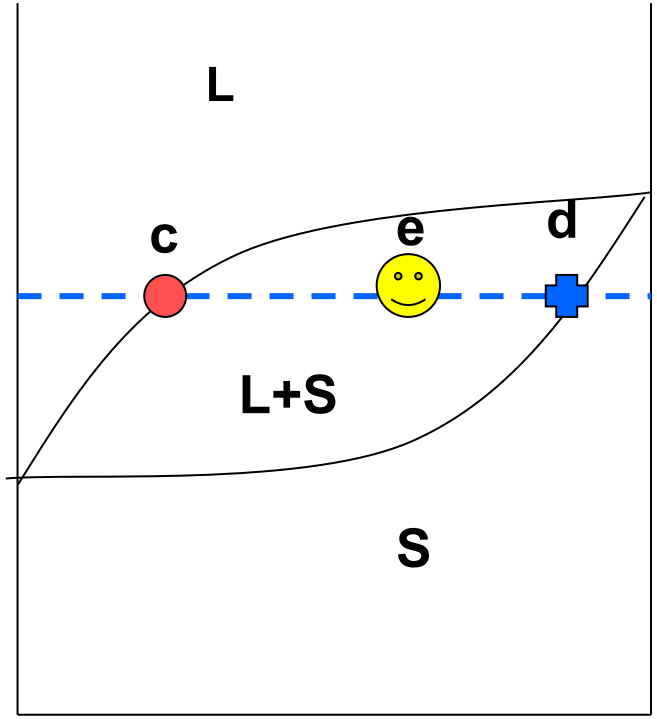
- Draw the tie line - - - - - -
- Individuate intersections with **liquidus** (•) and **solidus** (✚) lines
- The **composition of the liquid** is the value on the x axix correspondent to the intersection between the tie line and the liquidus line.
- The **composition of the solid** is the value on the x axis correspondent to the intersection of the tie line with the solidus line.

***LIQUIDUS CURVE*** = **above** it **only liquid** is present

***SOLIDUS CURVE*** = **below** it **only solid** is present

# Phases relative amounts determination in a biphasic zone, at equilibrium:

# LEVER RULE



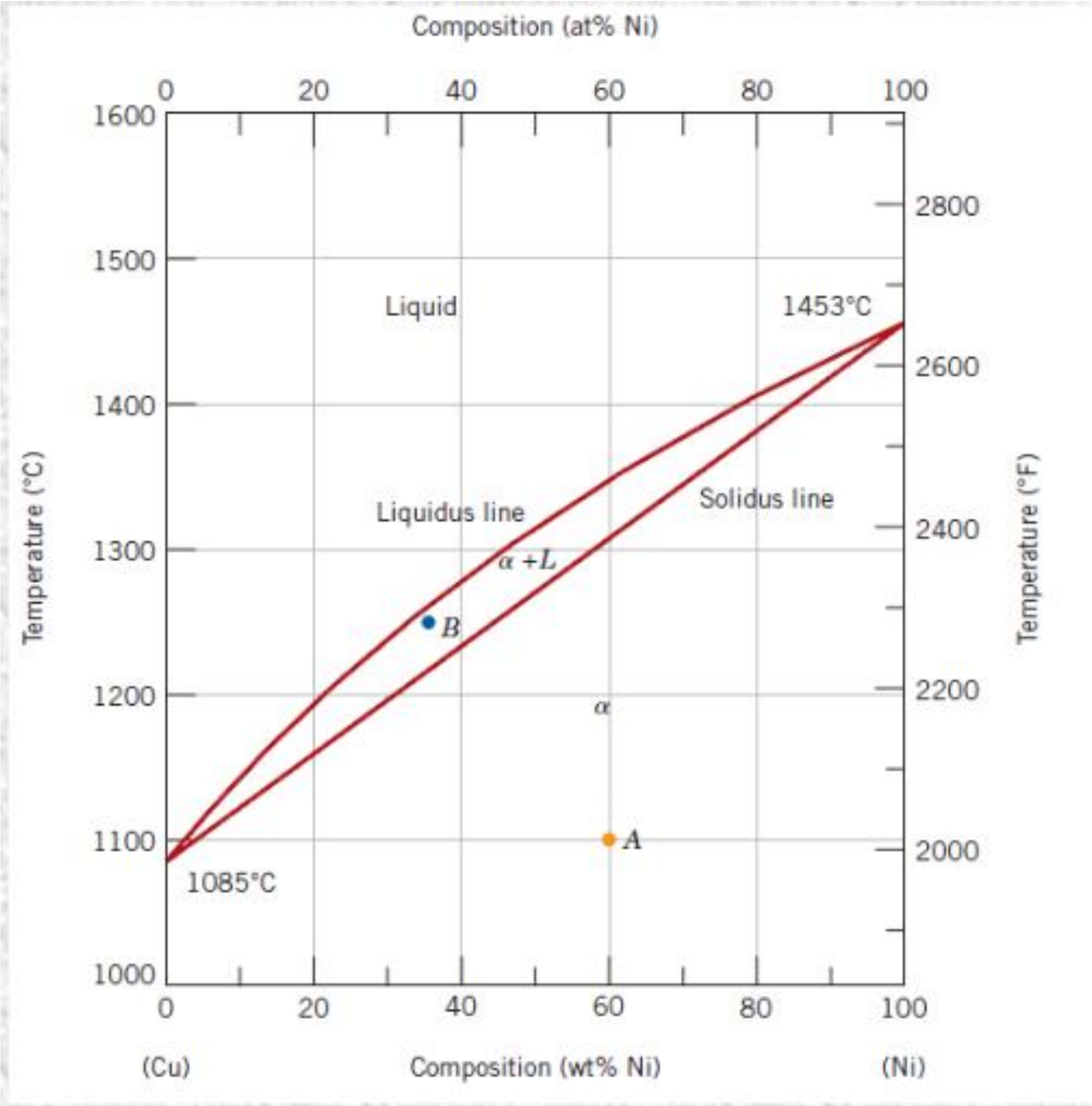

- Draw the tie line - - - - - -
- Individuate the intersection between tie line and liquidus line (● c) and solidus line (✚ **d**)
- The % of phases in a point of the biphasic region (e) is defined as:

**%L** = (ed/cd)*100

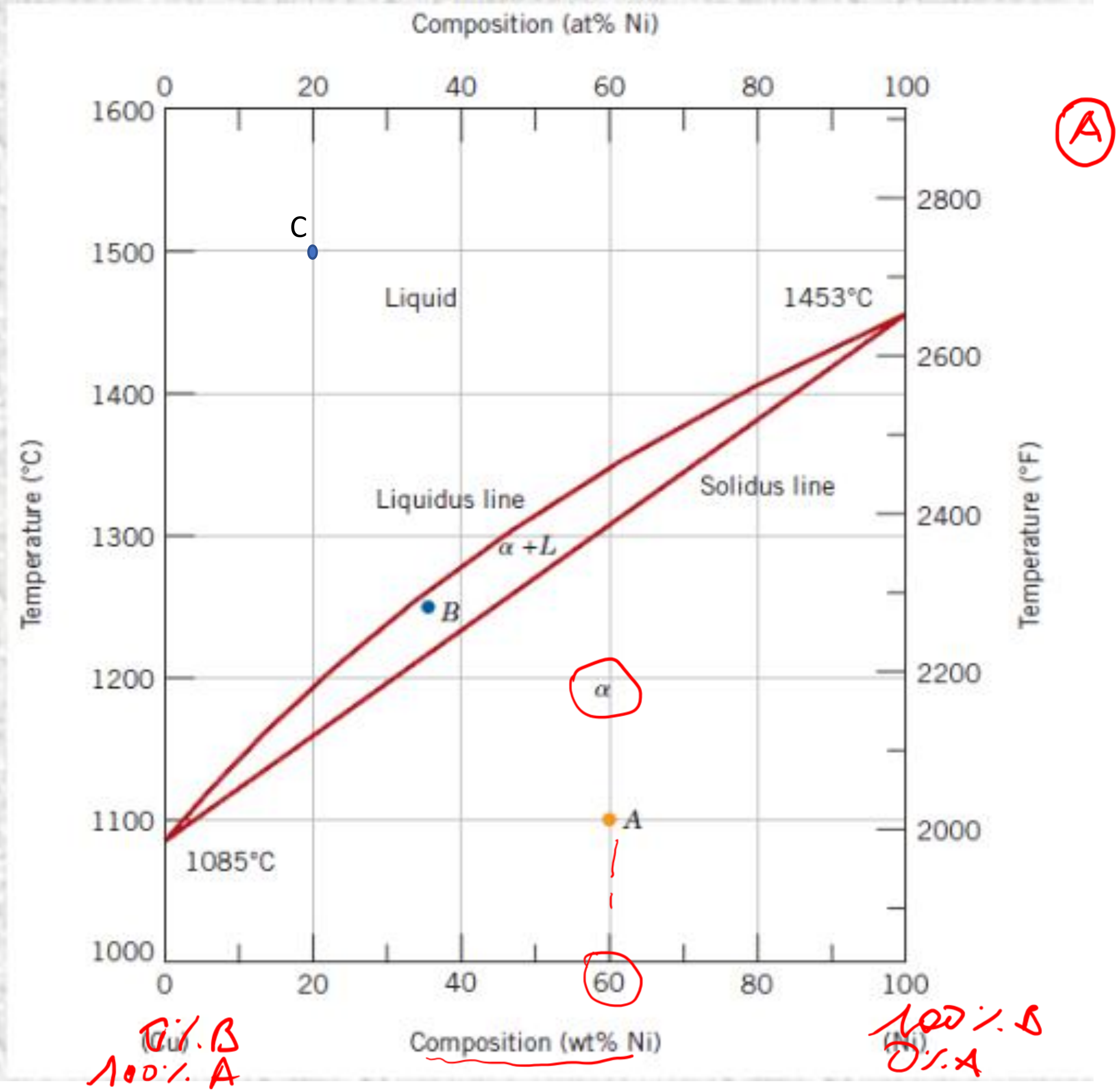
**%S** = (ce/cd)*100

**Exercise 1:** Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **complete solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

# Exercise 1: full solubility in the solid state

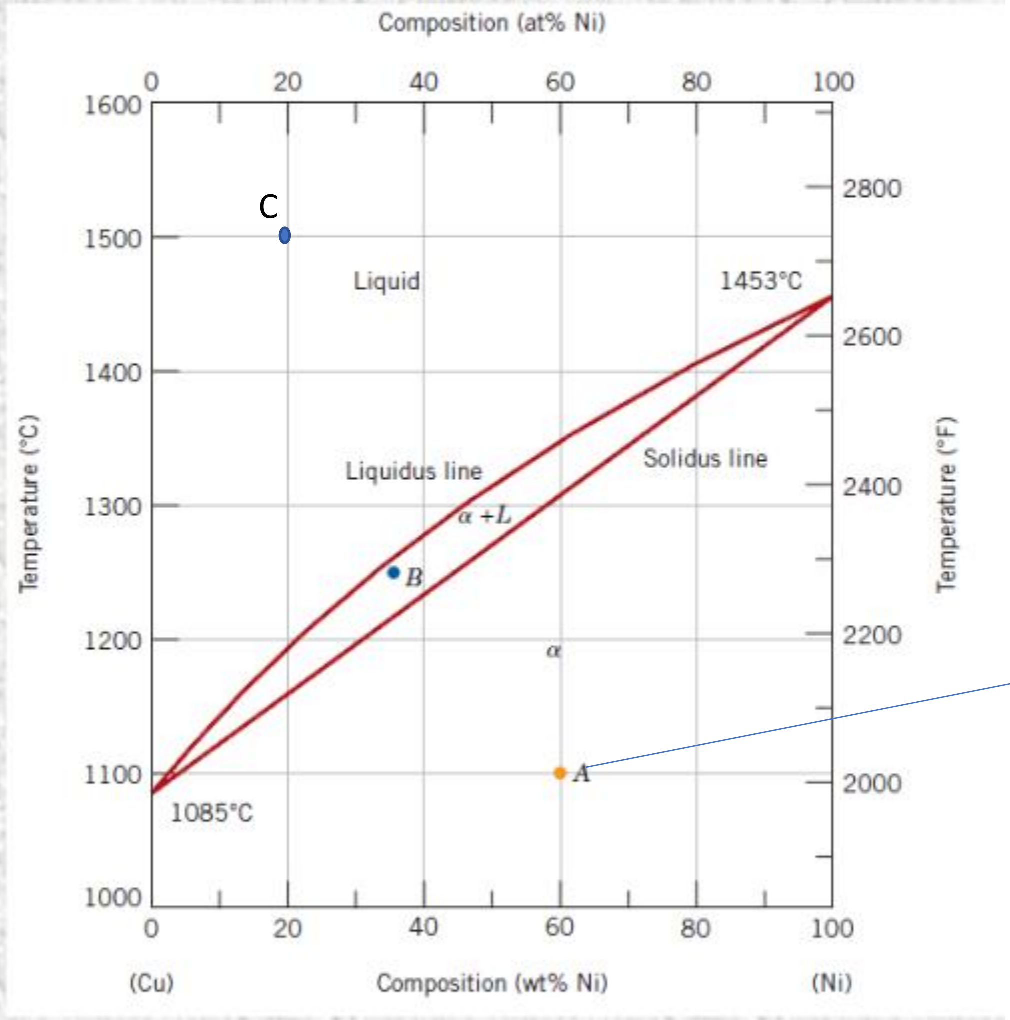


**Exercise 2**: determine the phases present, their composition and relative amounts in the point A, B and C indicated in the phase diagram. Schematize the microstructure in each point.



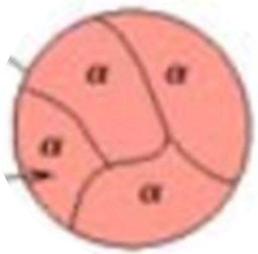

Ⓐ 60% Ni
40% Cu

Solid phase




## Point A:

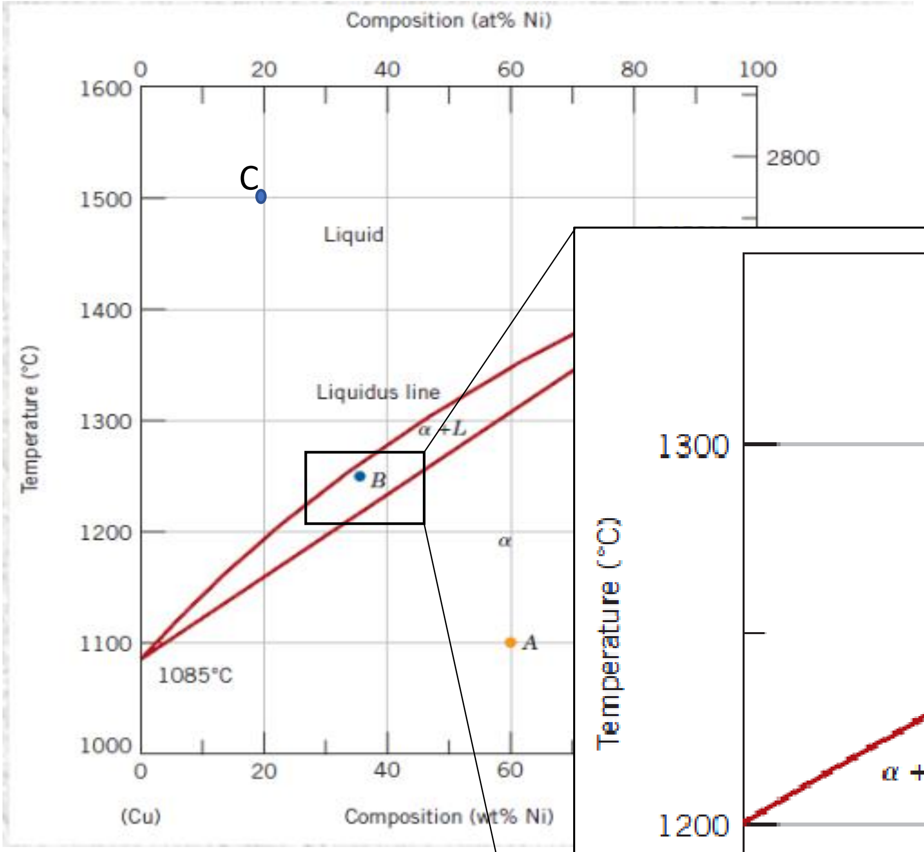
**Phases:** 1 phase solid
**Composition:** 60%Ni, 40% Cu
**Relative amount:** 100% solid phase

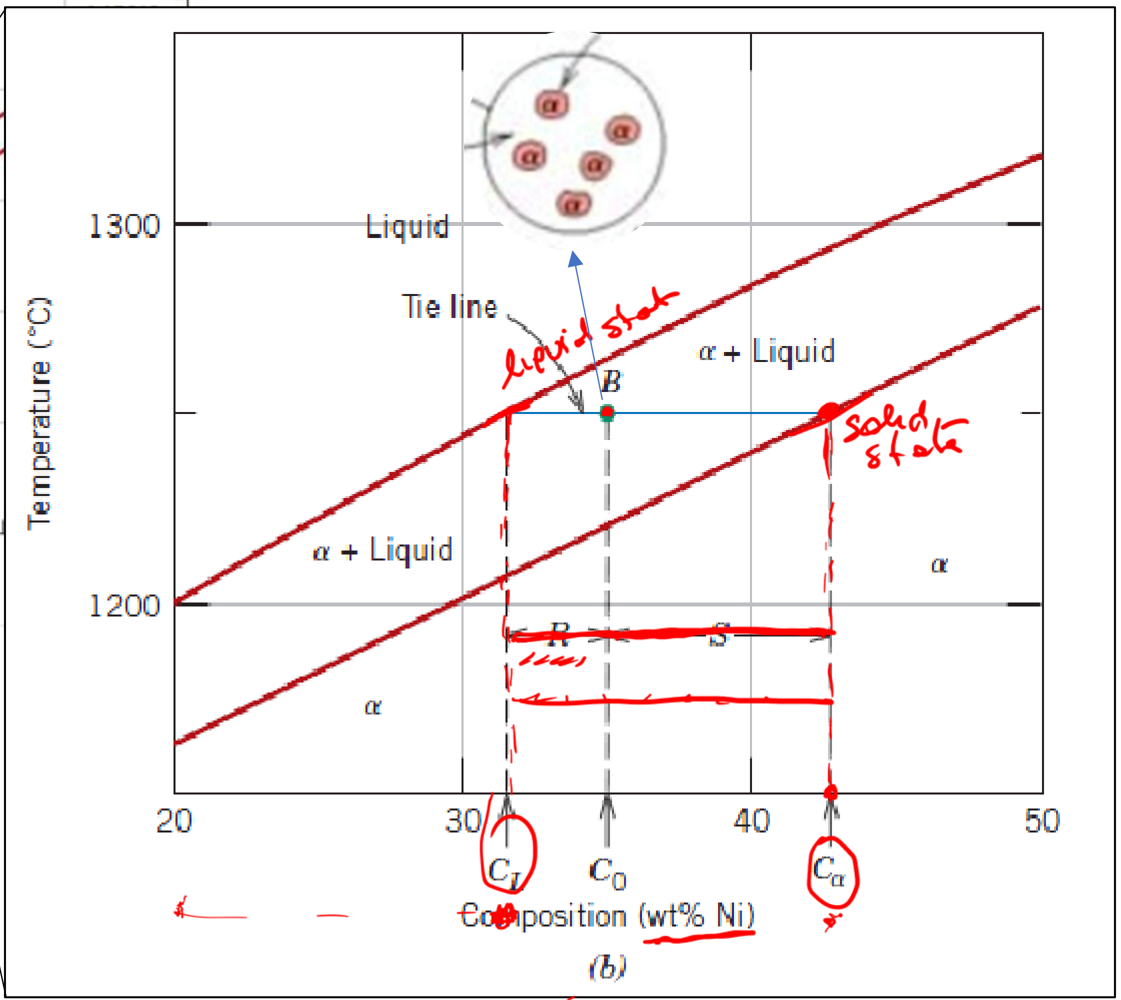


(B) Solid phases + liquid

$C_L = 32\%\ Ni;\quad 68\%\ Cu \rightarrow$ composition of comp A and B in the L

$C_\alpha = 43\%\ Ni;\quad 57\%\ Cu \rightarrow$ in S




**Point B:**

**Phases:** 2 pahses solid (α) +liquid (L)

**Composition**: 2 phases region → horizontal rule

**Composition of liquid (L):** CL= 32% Ni, 68%Cu

**Composition of solid (α):** C α= 43%Ni, 57% Cu

**Relative amount:** 2 phases region → lever rule

**% α**=R/(R+S)*100= (35-32)/(43-32)*100=27%

**%L**=S/(R+S)*100=(43-35)/(43-32)*100=73%

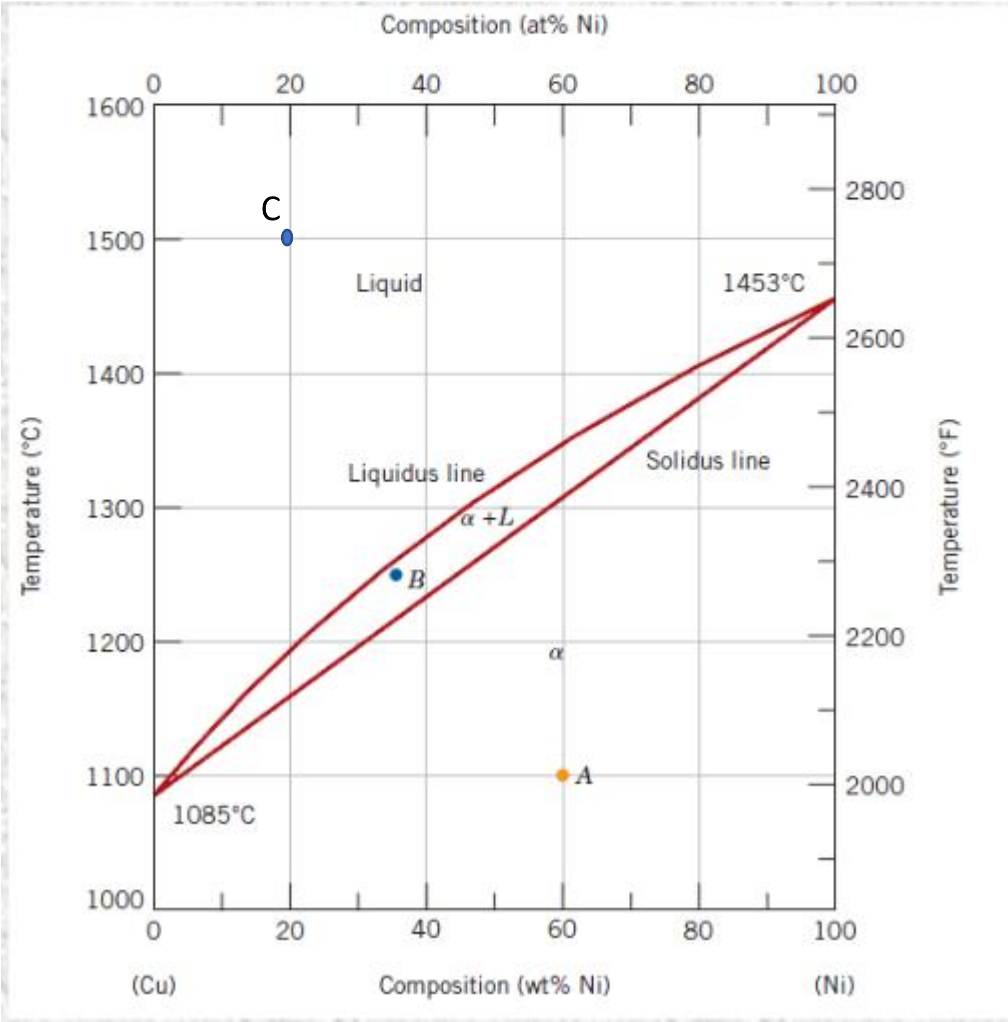
Relative amount

$L = S/R+S$

$S(\alpha) = R/R+S$

$L = (43-35/43-32) * 100 = 73\%$

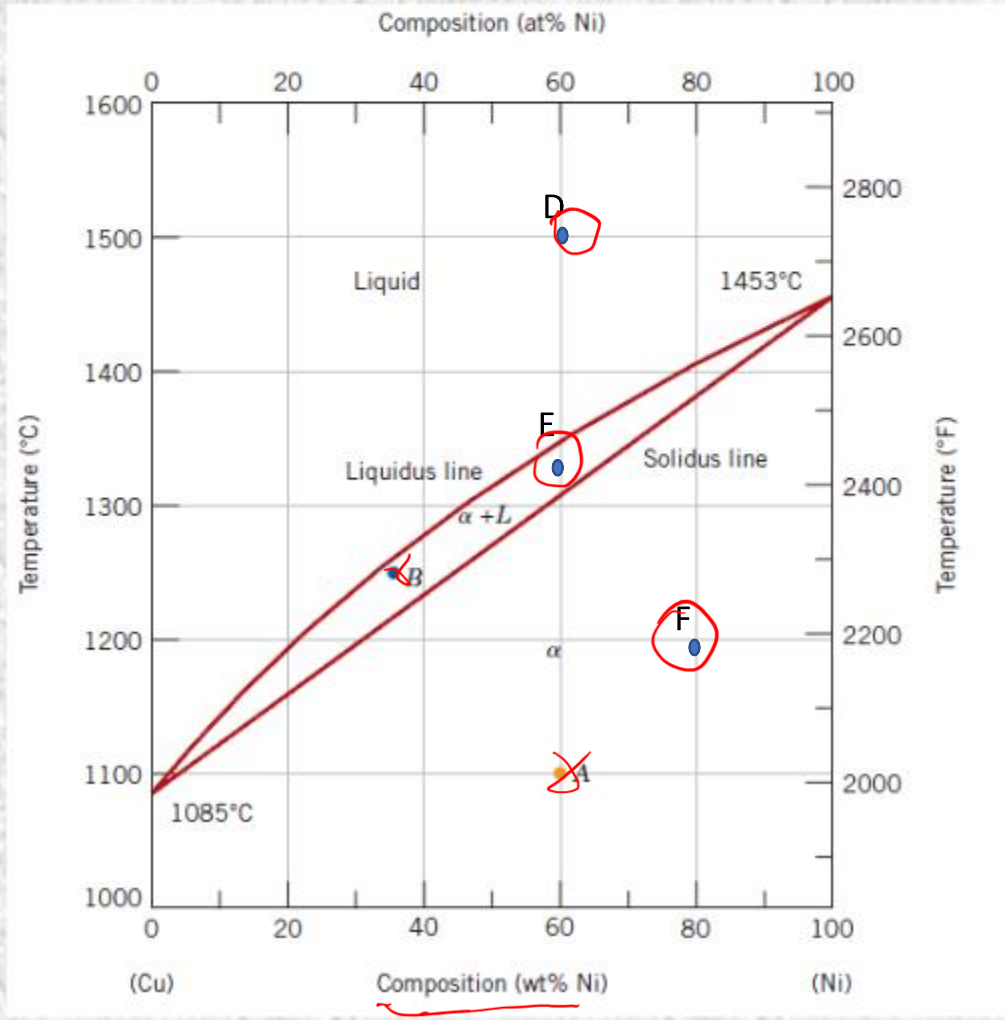
$S = (35-32/43-32) * 100 = 27\%$

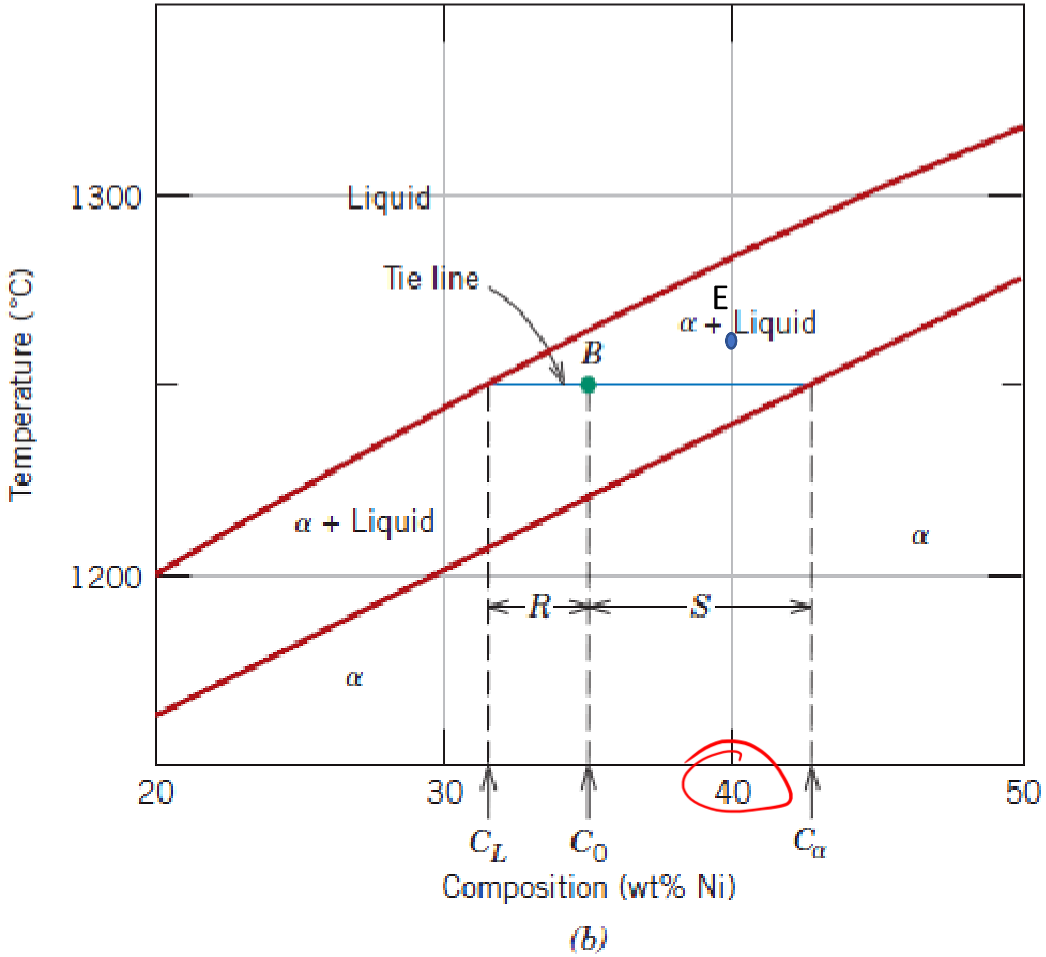


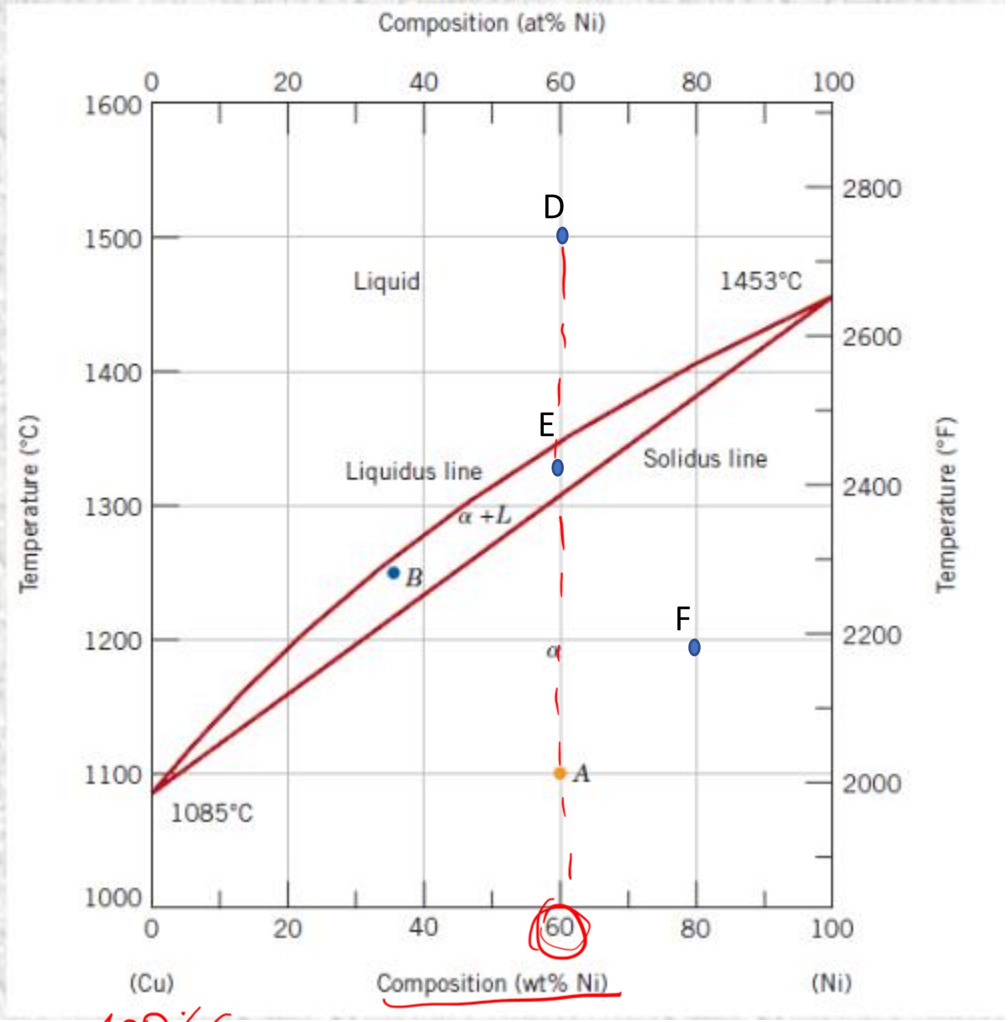

**Point C:**

**Phases:** 1 phase liquid
**Composition:** 20%Ni, 80% Cu
**Relative amount:** 100% liquid phase

**Exercise 3:** determine the phase present, their composition and relative amounts in the point D, E and F indicated in the phase diagram. Schematize the microstructure in each point.



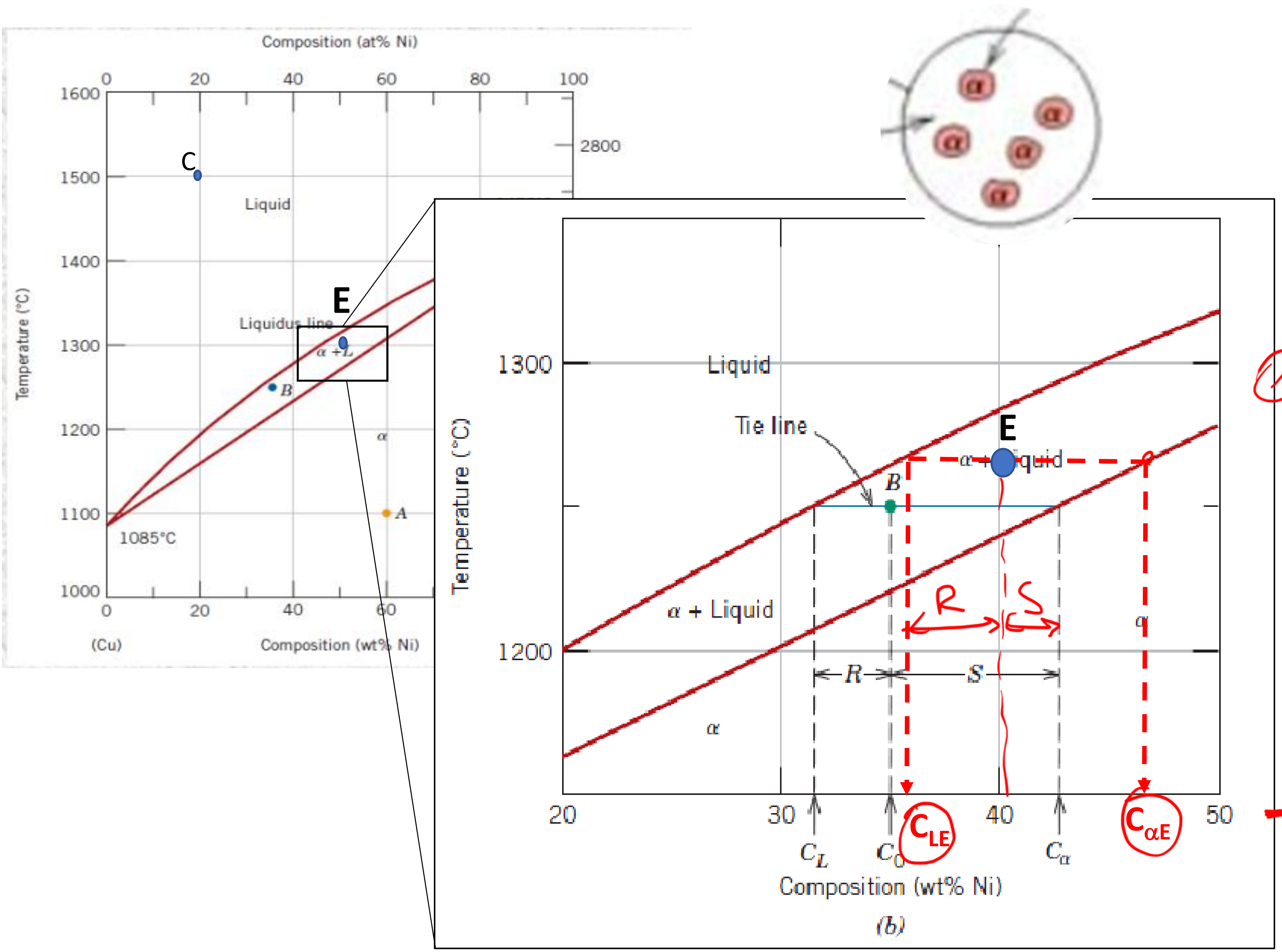



100% Cu . . . . . . .

(100 − 60) = 40 %

## Point D:

**Phases:** 1 phase liquid
**Composition:** 60%Ni, 40% Cu
**Relative amount:** 100% liquid phase




**Point E:**

**Phases:** 2 phases =solid (α) +liquid (L)

**Composition**: 2 phases region → horizontal rule

(1) **Composition of liquid (L):**

$C_{LE}$= 36% Ni, 64%Cu

**Composition of solid (α):**

C αE= 46%Ni, 54% Cu

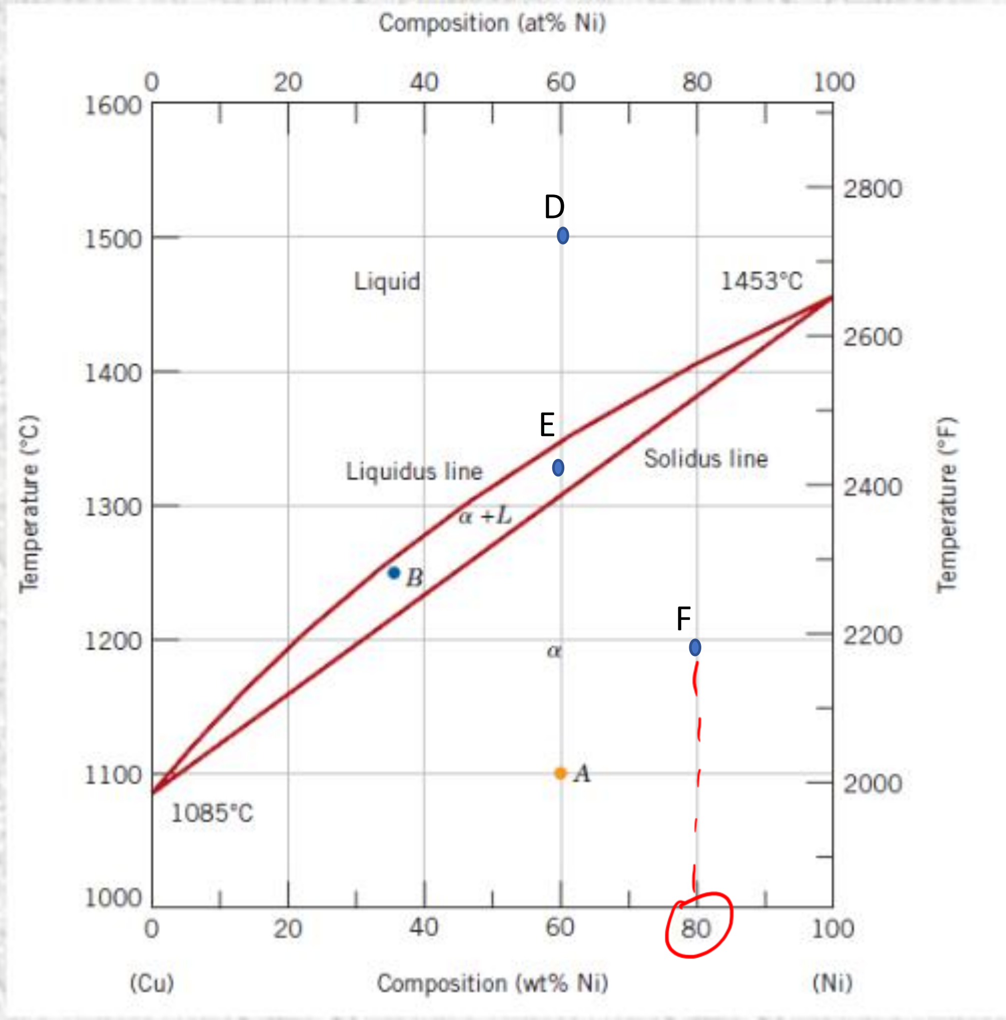
**Relative amount :** 2 phases region → lever rule

R / R+S

**% α**= (40-36)/(46-36)*100=40%

**%L**=(46-40)/(46-36)*100=60%

S / R+S

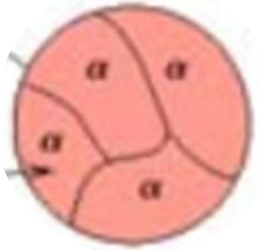



**Point F:**

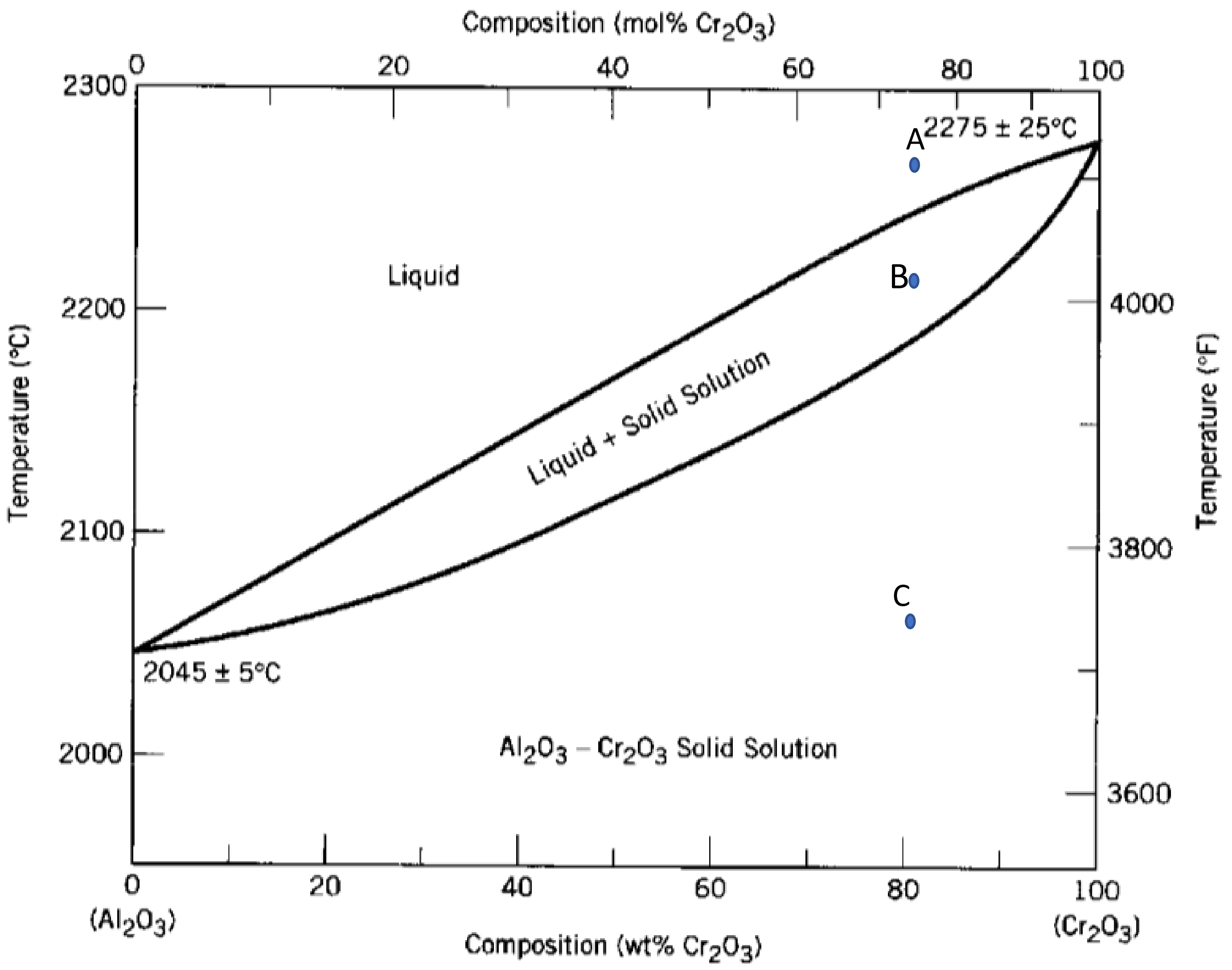
**Phases:** 1 phase solid

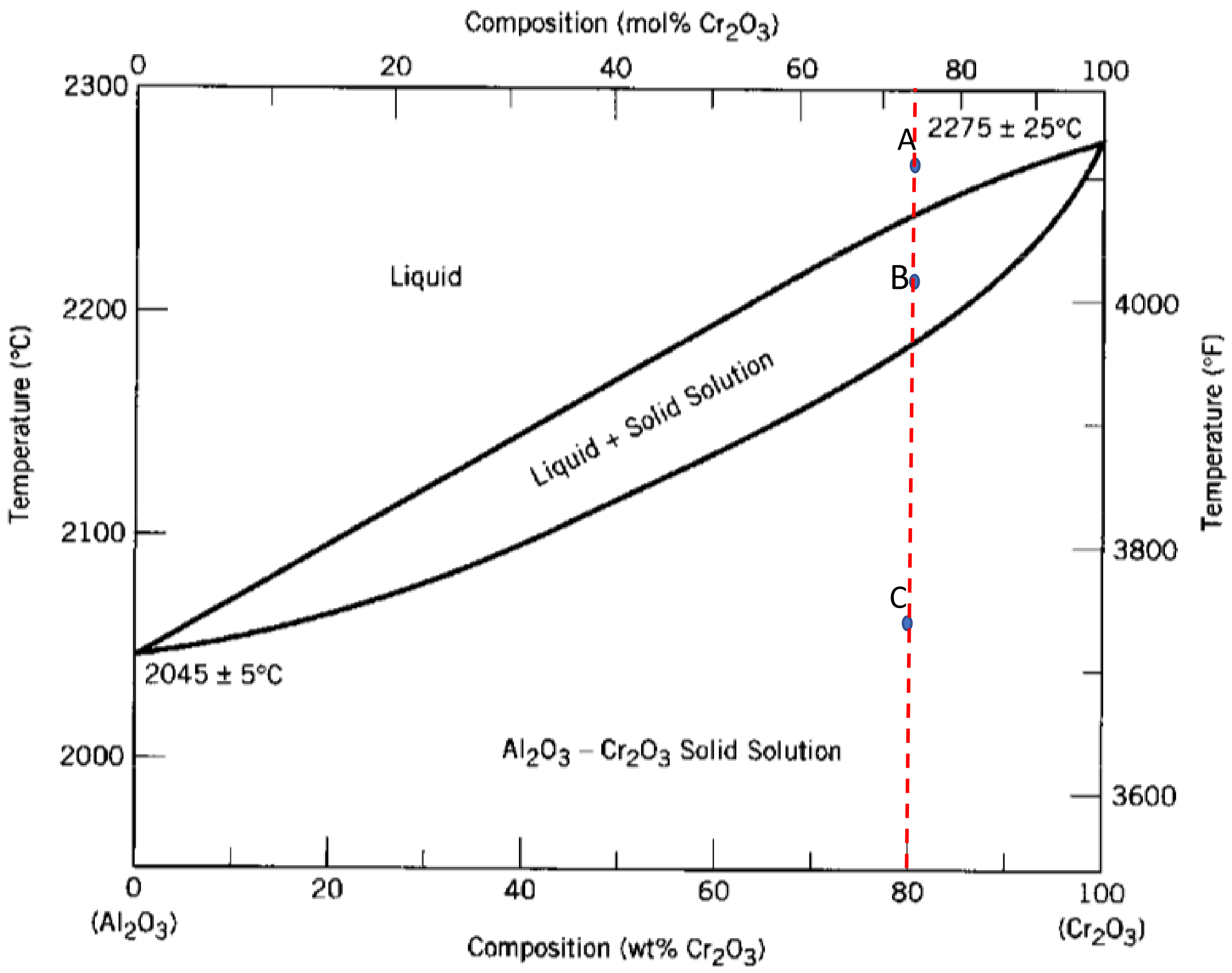
**Composition:** 80%Ni, 20% Cu

**Relative amount:** 100% solid phase



**Exercise 4:** determine the phase present, their composition and relative amounts in the point A, B and C indicated in the phase diagram.



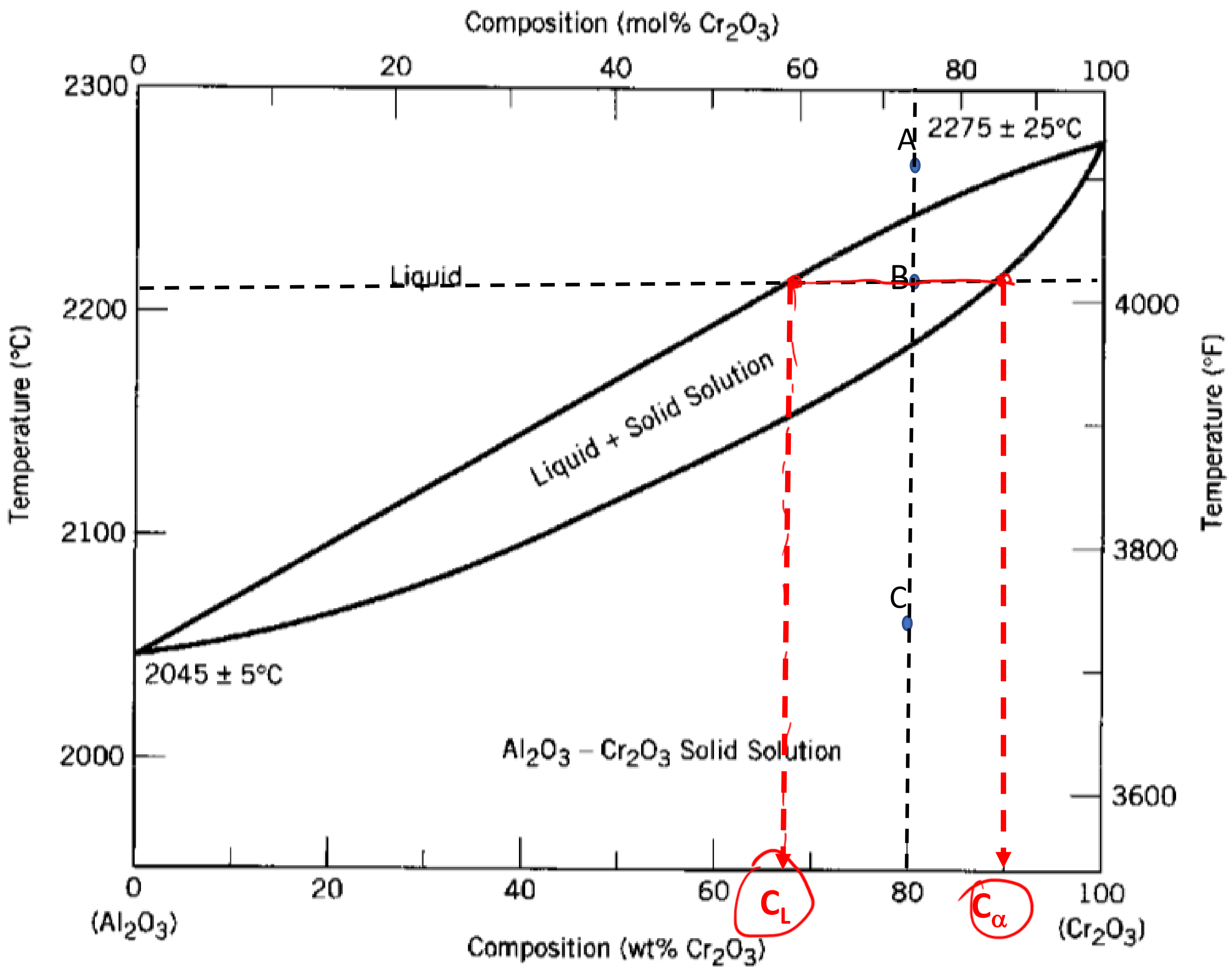



**<u>Point A:</u>**

**Phases:** 1 phase liquid

**Composition:** 80%$Cr_2O_3$, 20% $Al_2O_3$

**Relative amount:** 100% liquid phase




**Point B:**

**Phases:** 2 phases solid ($\alpha$) +liquid (L)

**Composition**: 2 phases region → <u>horizontal rule</u>

**Composition of liquid (L):**

$C_L$= 68% $Cr_2O_3$, 32% $Al_2O_3$

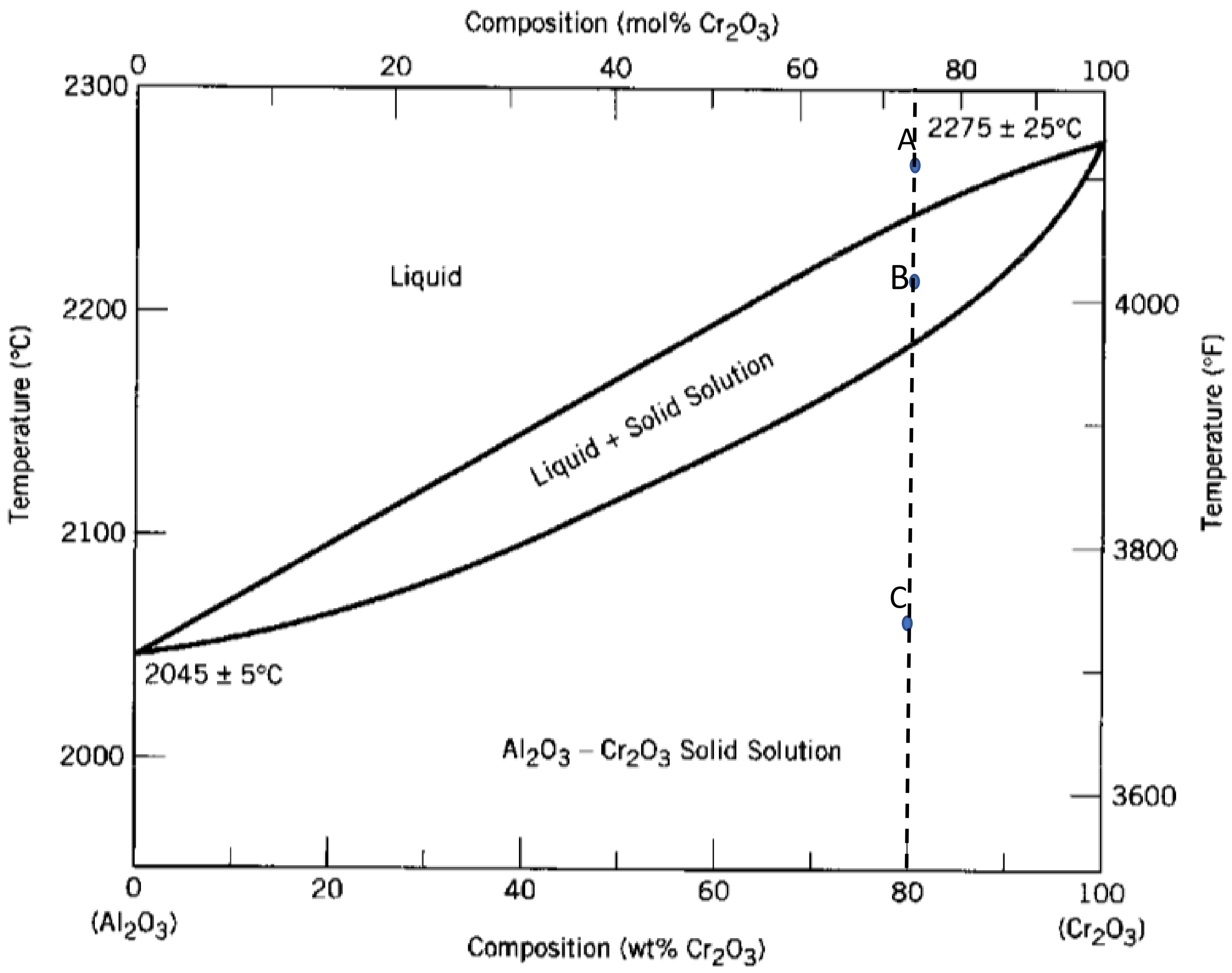
**Composition of solid ($\alpha$):**

$C\alpha$= 90% $Cr_2O_3$, 10% $Al_2O_3$

**Relative amount:** 2 phases region → <u>lever rule</u>

**% $\alpha$**= (80-68)/(90-68)*100=54.5%

**%L**=(90-80)/(90-68)*100=45.5%

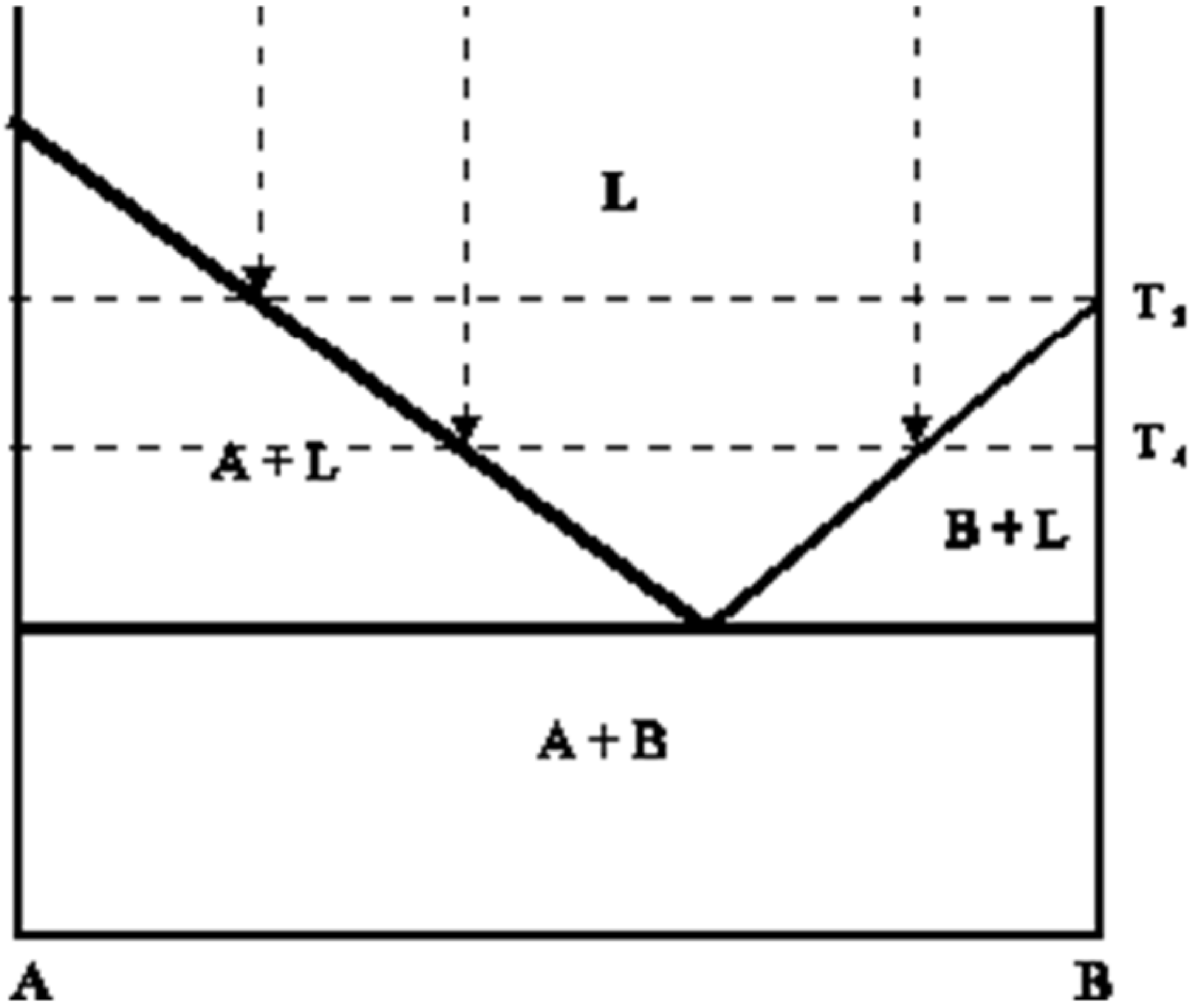



**Point C:**

**Phases:** 1 phase solid
**Composition:** 80%$Cr_2O_3$, 20% $Al_2O_3$
**Relative amount:** 100% solid phase

**Exercise 5:** Draw a phase diagram (for generic components A and B), with **complete solubility in the liquid state** and **no solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

# Complete solubility in the liquid state, no solubility in the solid state




L = liquid Region

A+L = Solid phase of A + liquid rich B

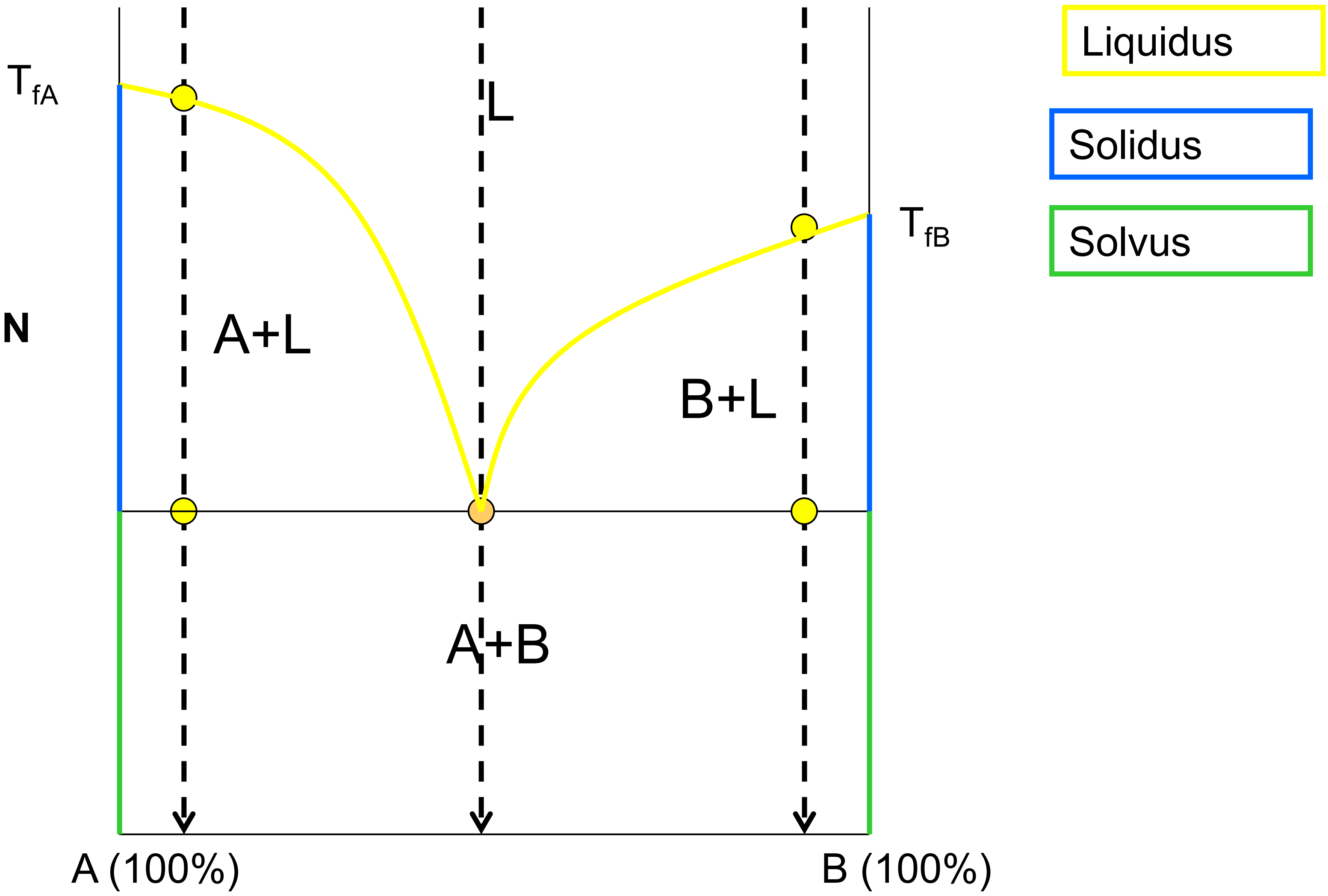
B+L = solid phase of B liq rich A

Eutectic → L → A+B

# COMPLETE SOLUBILITY IN THE LIQUID STATE AND NO SOLUBILITY IN THE SOLID STATE, EUTECTIC

**EUTECTIC TRANSFORMATION**

**$L \leftrightarrow A+B$**



Exercise 6: determine the phases present, their compositions and relative amounts in the point X.

X

phases : Solid (A) + L

100 % $C_A$

$C_L$ = 45 % B

55 % A

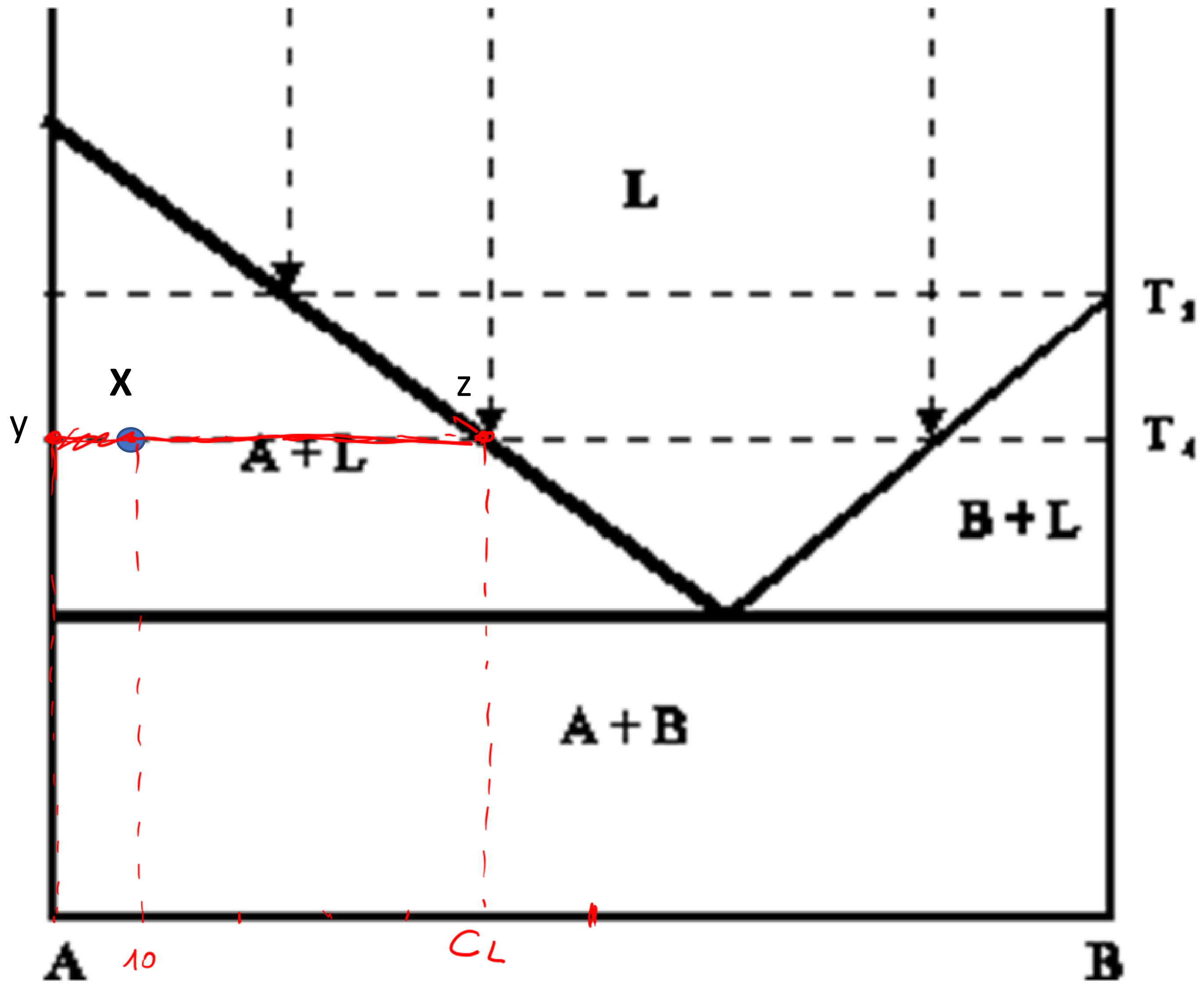
$\% A = \frac{xz}{yz} \times 100$

$= \frac{45 - 10}{45} \times 100$

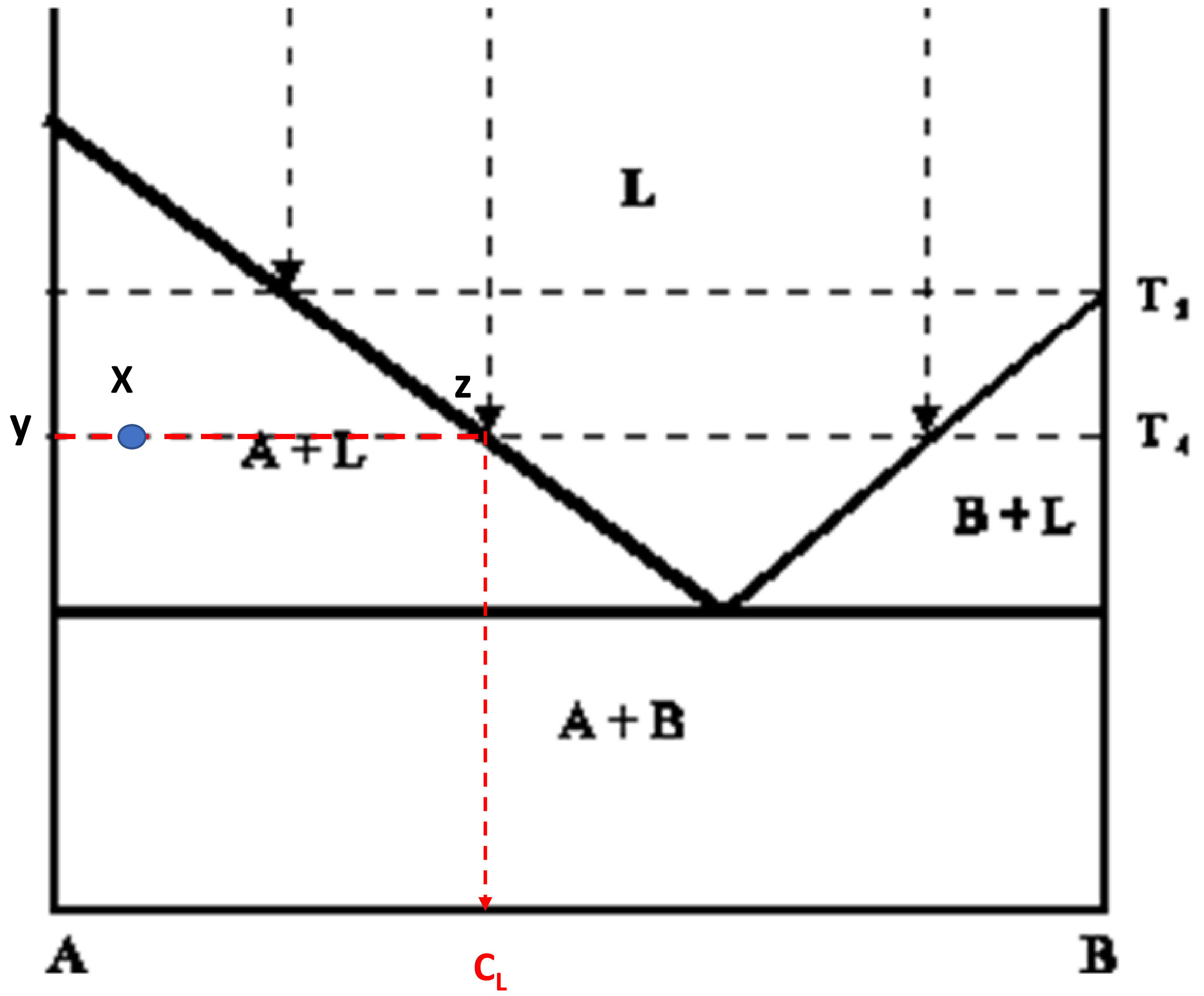
$= 78 \%$

$\% L = \frac{yx}{yz} = \frac{10}{45}$

$= 22 \%$



**Exercise 6:** determine the phases present, their compositions and relative amounts in the point X.




__**Point X**__

**Phases:** solid A+ Liquid L

**Compositions:** 2 phases region → horizontal rule→intersection with each phase region and draw until the axis→ you find:

$C_A$: 100% A 0%B

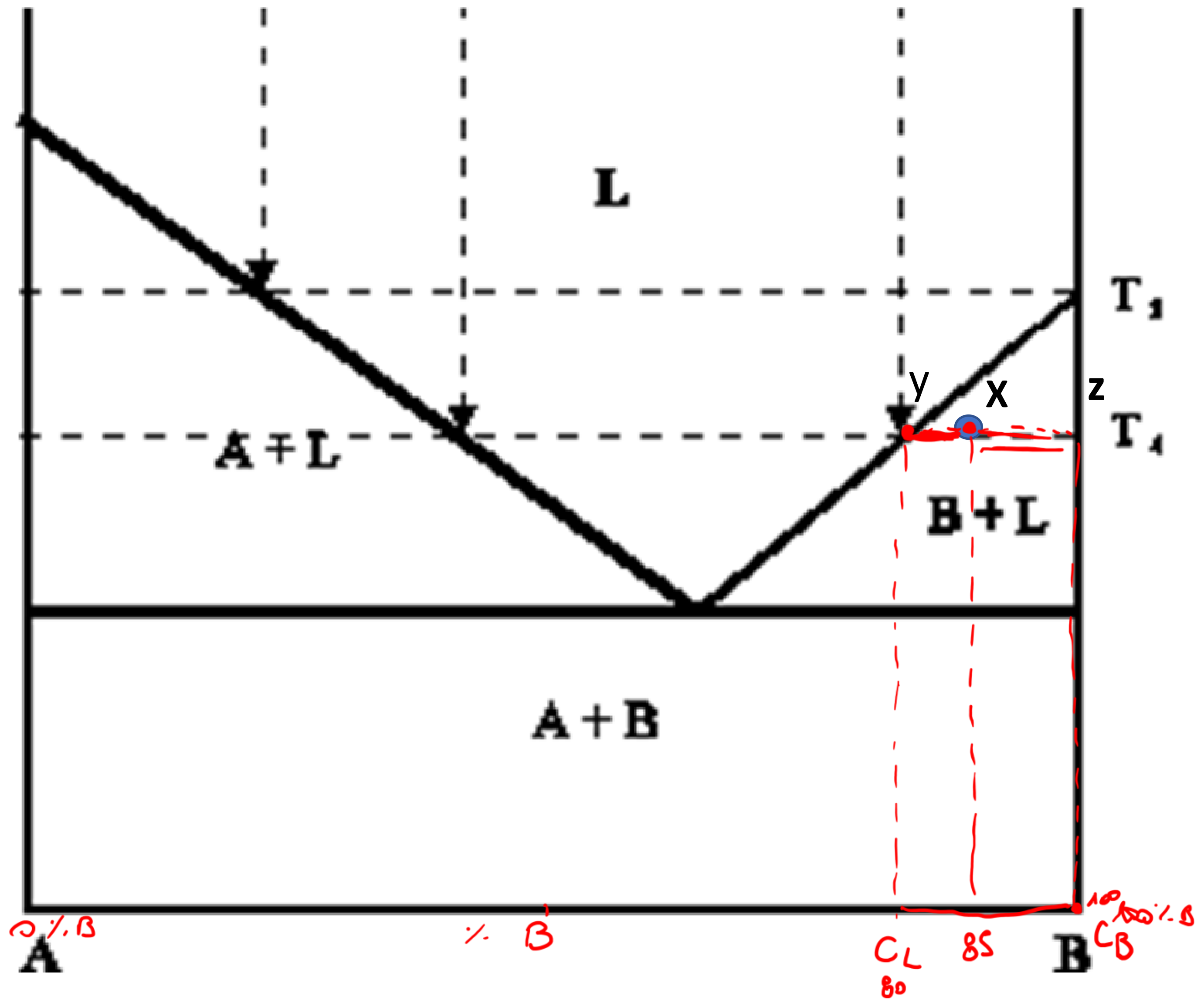
$C_L$= 45%B, 55%A

**Relative amounts**

%A= (Xz)/(yz)*100= (45-10)/(45-0)*100=78%

%L=(yX)/(yz)=(10-0)/(45-0)*100=22%

**Exercise 7:** determine the phases present, their compositions and relative amounts in the point X.




2 phases: Solid B + Liq phase

Horizontal Rule

$C_L$ = 80% B 20% A

$C_B$ = 100% B 0% A

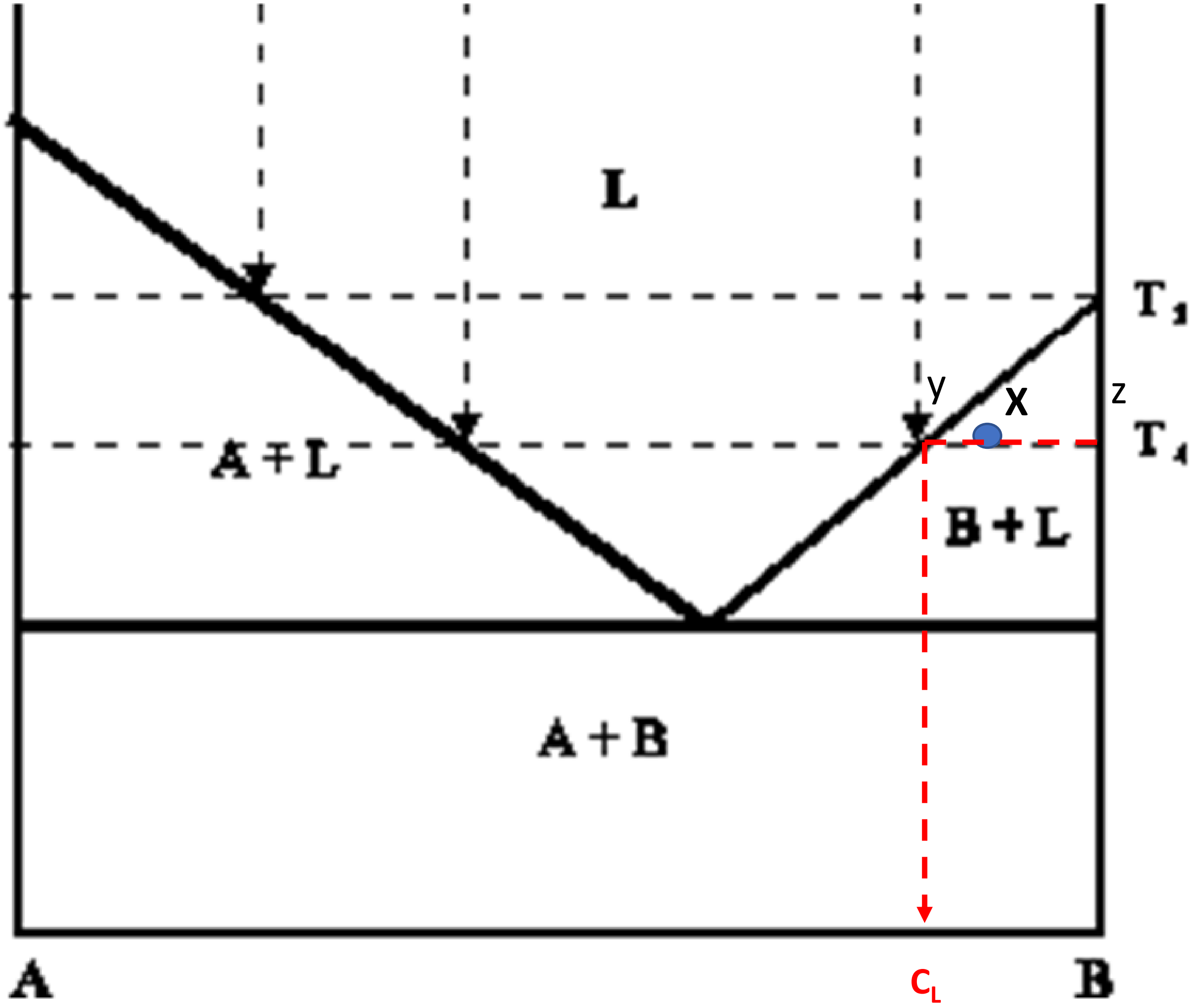
Lever Rule

$\%B = xy/yz * 100 = \frac{85-80}{100-80} \times 100 = 25\%$

$\%L = xz/yz * 100 = 100-85/100-80 * 100\%$

$= 75\%$

**Exercise 7:** determine the phases present, their compositions and relative amounts in the point X.




**Point X**

**Phases:** solid B+ Liquid L

**Compositions:** 2 phases region → horizontal rule

$C_B$: 100% B

$C_L$= 80%B, 20%A

**Relative amounts**

%B= (Xy)/(yz)*100= (85-80)/(100-80)*100=25%

%L=(Xz)/(yz)=(100-85)/(100-80)*100=75%